

Subject: Client Side Scripting	Subject Code: 316005
Semester: 06	Course: CO6K-A
Laboratory No: L001A	Name of Subject Teacher: Prof. Sneha Patange
Name of Student: Shreyas Purushottam Bandekar	Roll ID: 23203A0069

Experiment No:	17
Title of Experiment:	Write a program to implement states of React Hooks.

I. Practical Significance:

Implementing states using React Hooks helps developers manage and update dynamic data in functional components without using class components. It allows components to remember values, respond to user actions, and update the UI automatically when data changes.

➤ Code

- Write a React program to change the text color when a button is clicked using the `useState` hook.

Code:-

App.jsx:

```
import React from "react";
import ReactDOM from "react-dom/client";
import { useState } from "react";
const App = () => {
  const [color, setColor] = useState("red");
  const changeColor = () => {
    setColor("blue");
  };
  return (
    <div>
      <h1 style={{ color: color }}>My favourite colour is Red!</h1>
      <button onClick={changeColor}>Set Blue</button>
    </div>
  );
};
```

```
};  
export default App
```

index.jsx:

```
import React from 'react'  
import ReactDOM from 'react-dom/client'  
import App from './App.jsx'  
  
ReactDOM.createRoot(document.getElementById('root')).render(  
  <React.StrictMode>  
    <App />  
  </React.StrictMode>  
)
```

Output:-

My favourite colour is Red!

Set Blue

My favourite colour is Red!

Set Blue

- Write a ReactJS program to implement a counter using the useState Hook and update the counter value on button click.

Code:-

App.jsx:

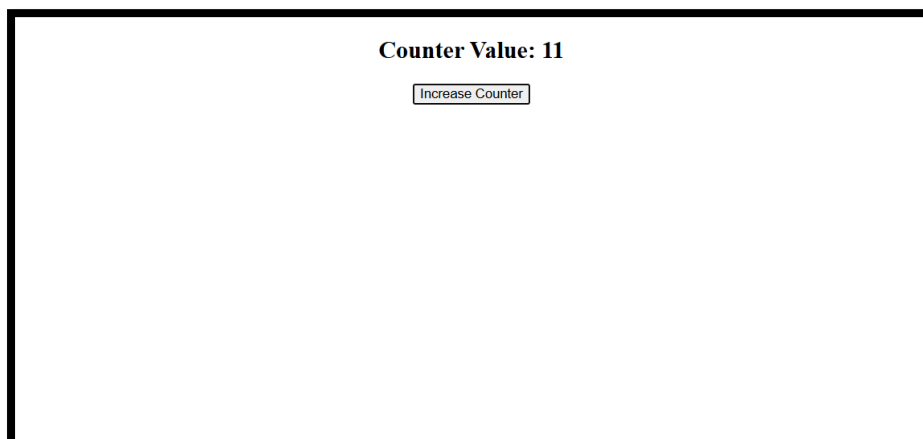
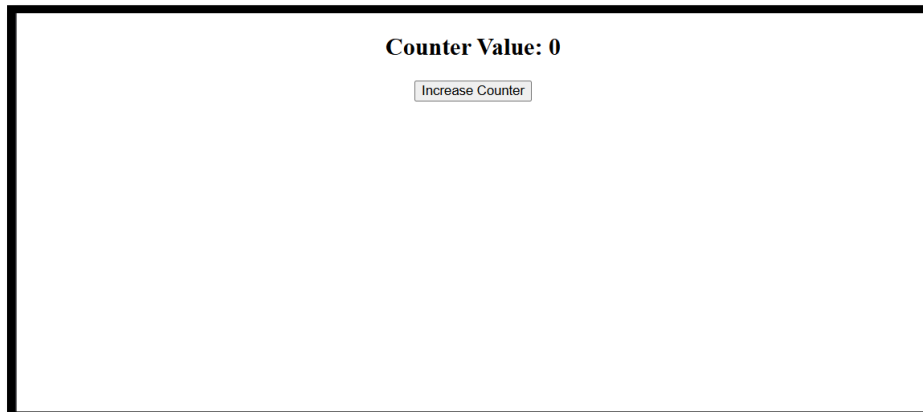
```
import React, { useState } from "react";  
import ReactDOM from "react-dom/client";  
function App() {  
  const [count, setCount] = useState(0);  
  // Function to update counter  
  const updateCounter = () => {
```

```
    setCount(count + 1);  
  };  
  return (  
    <div style={{ textAlign: "center" }}>  
      <h2>Counter Value: {count}</h2>  
      <button onClick={updateCounter}>  
        Increase Counter  
      </button>  
    </div>  
  );  
}  
export default App;
```

index.jsx:

```
import React from 'react'  
import ReactDOM from 'react-dom/client'  
import App from './App.jsx'  
  
ReactDOM.createRoot(document.getElementById('root')).render(  
  <React.StrictMode>  
    <App />  
  </React.StrictMode>,  
)
```

Output:-



II. Practical Related Questions:

1. Write a React program to change the background color of a div when a button is clicked using the useState hook.

Ans:

Code:-

App.jsx:

```
import React from "react";
import ReactDOM from "react-dom/client";
import { useState } from "react";
const App = () => {
  const [color, setColor] = useState("Red");
  function changeColor() {
    setColor("Blue");
  }
  return (
    <div style={{ textAlign: "center", marginTop: "40px" }}>
      <div style={{ backgroundColor: color, padding: "50px" }}>
        <h1>My favourite colour is {color}!</h1>
      </div>
      <br />
      <button onClick={changeColor}>Set Blue</button>
    </div>
  );
};
export default App;
```

index.jsx:

```
import React from 'react'
import ReactDOM from 'react-dom/client'
import App from './App.jsx'

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>,
)
```

Output:-

My favourite colour is Red!

Set Blue

My favourite colour is Blue!

Set Blue

2. Write a React program to manage user input using the useState hook.

Ans:-

Code:-

App.jsx:

```
import React, { useState } from "react";
import ReactDOM from "react-dom/client";
function UserInput() {
  // State to store user input
  const [name, setName] = useState("");
  return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h2>React User Input Example</h2>
      <input
        type="text"
        placeholder="Enter your name"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <h3>Hello {name}</h3>
    </div>
```

```
);  
}  
export default UserInput
```

index.jsx:

```
import React from 'react'  
import ReactDOM from 'react-dom/client'  
import UserInput from './App.jsx'  
  
ReactDOM.createRoot(document.getElementById('root')).render(  
  <React.StrictMode>  
    <UserInput />  
  </React.StrictMode>,  
)
```

Output:

React User Input Example

Hello

React User Input Example

Hello Vidyalankar

III. Exercise:**1. What is the purpose of the useState() hook in React?****Ans:**

The useState() hook is used to **create and manage state in functional components** in React. It allows components to store and update data such as user input, counters, or form values. When the state changes, React automatically **re-renders the component** to display the updated data.

2. Why do we use a setter function like setCount in React Hooks?**Ans:**

The setter function (such as setCount) is used to **update the state value** created by useState(). React requires using this setter function instead of directly changing the state so that it can **detect the change and re-render the component** properly.

IV. Conclusion:

This practical demonstrates how React Hooks can be used to manage and update state in functional components. It helps in understanding how dynamic data changes based on user actions and how React automatically updates the user interface when the state changes.

Marks Obtained			Dated Signature of Teacher
Process Related (15)	Product Related (10)	Total (25)	